Project 1 Written Report

ME 370

9/18/26

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1. DESIGN CONSIDERATIONS

1.1 MECHANISM

We are mainly considering using a crank-rocker mechanism for our project's primary mechanism. During the lab, we noticed the crank-slider jamming and saw the added complexity needed for this mechanism to move smoothly. Because of this, we decided to focus our mechanical design around a simple 4-bar mechanism without a crank-slider.

1.2 MATERIALS

Some of the materials we used for structures and links during Lab 1 were acrylic and plastic (3D filament). The acrylic provided more strength at a smaller thickness, and the smooth surface finish reduced friction when linkages passed each. This realization changes our material choice towards acrylic, although 3D-printed surfaces are smooth enough for the rapid prototyping process.

1.3 FASTENING

Fasteners used in this lab included nuts and bolts, shoulder bolts, barrel fasteners, heat set inserts, and bearings. We noticed the ease of installation and reduced friction with the barrel fastens, and we liked the compactness of the heat set inserts in both acrylic and 3D-filament. The smooth contacting surfaces of barrel fasteners and the stress distribution of heat set inserts makes these two fasteners our top choices.

1.4 SHAFT AND PIN SUPPORT

This lab taught us the importance of shaft and pin support when making reliable designs. The geartrain assembly and the slider and crank assemblies showed us how less support on shafts transferring forces causes inefficiencies and binding. Making sure our input shaft is properly supported and constrained will help transfer forces from the crank to the legs, resulting in smoother function and less exertion.

1.5 JOINERY

This lab taught us the importance of general joinery when designing structures. We saw in Lab 1 that designs that have rigidity transfer forces more efficiently, maintain consistent functionality, and are less prone to wear. Designing our mechanism with a strong, rigid backbone will help our leg linkages maintain a consistent path, stay aligned to prevent bending, and improve force delivery. We will also consider adding small slots instead of holes in links for alignment adjustability.

1.6 MANUFACTURING TOLERANCES

Manufacturing tolerances are important when designing mechanisms with motion. We learned in the lab how small out-of-tolerances in mechanisms such as the slider crank can cause binding, bending, or excessive friction. When trying to screw in the link to the sets of spaced heat inserts, we found ourselves stressing the link to fit on different sets. Going into future designs, we plan on keeping tighter tolerances on pivot joints and hole alignments and looser tolerances on non-critical frame or link distances.

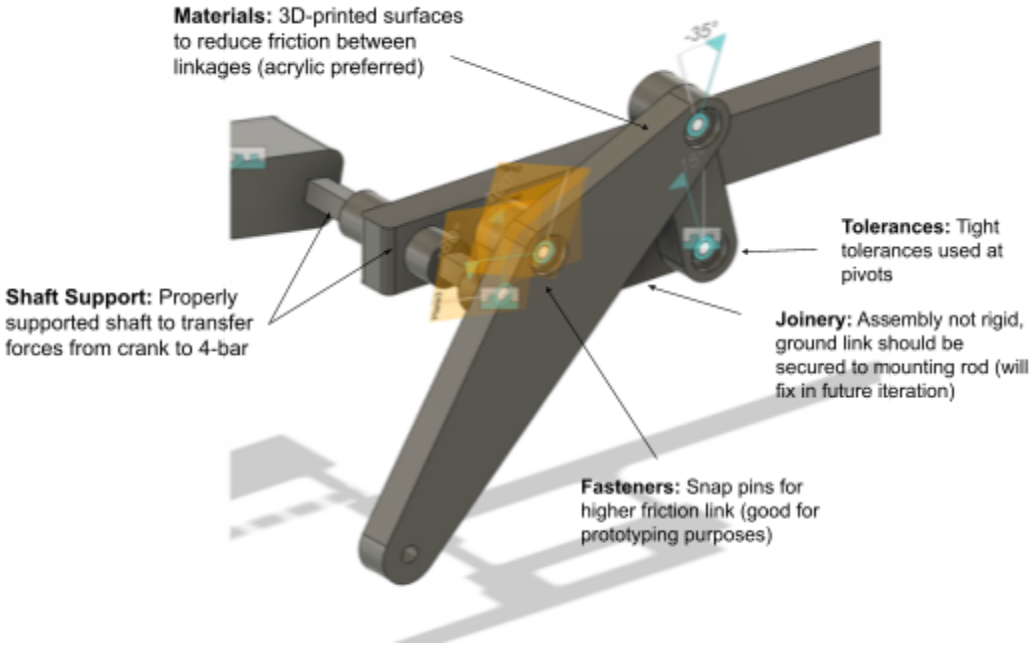


Fig 1. Moderate-fidelity prototype with annotated design considerations

2. LOW-FIDELITY PROTOTYPING LAB

2.1 DESIGN VARIABLES AND CONSTRAINTS

Design Variables:	**Constraints:**
• Mechanism topology • Link lengths/ pivot locations • Foot placement, shape, material • Foot path (stride length/height) • Number and phasing of legs	• Continuous contact • Generate 2N of force • Operate from single hand crank • 24 x 24 x 24 cm size • Table-mounting capability • $60 budget

2.2 DESIGN QUESTIONS AND METHODS

Q1) How does changing the linkage lengths and pivot positions change the trajectory that the foot takes?

Method: Built 2 hand-cranked four-bar linkages made from popsicle sticks and cardboard, with varied linkage lengths and pivot locations. A six-bar linkage and a crank-slider were also considered but only sketched, since they were hard to build at low-fidelity.

Information provided: Both configurations gave a closed foot path, with a longer crank providing a longer stride length. The trajectory changed from a long oval shape to a stubby claw shape. Because of the wide range of usable paths with the fewest parts, we prioritised four-bar mechanisms.

Why suitable: Turning a cardboard linkage by hand shows the whole trajectory at once, and the parts are quick to re-pin, so we compared many configurations in one session.

Q2) Does the mechanism stay within the 24 × 24 × 24 cm envelope through its full range of motion?

Method: We turned both physical four-bars through a full revolution inside a 24 cm square outline, tracing the outermost path of every link. The six-bar and crank-slider were checked on scaled sketches by drawing their extreme positions inside the same outline.

Information provided: Both four-bars stayed comfortably inside the envelope. The six-bar's extra links swept a larger outline and the crank-slider required a fixed guide rail that took up static space. The key finding was that the sweep envelope determines size, not just the resting position.

Why suitable: Tracing the motion against a full-scale outline shows the swept extent directly, which a static model or a single sketch position cannot.

Q3) Can mechanisms such as a geartrain be incorporated to create a simple two legged machine?

Method: We created a sketch of three gears to explore whether we could create multiple legged walkers. Multiple legs were considered with multiple gears but we chose not to make a physical prototype due to the size constraint and complexity.

Information provided: Sketch uses a central input gear to turn two outer gears that are 180 degrees out of phase. The gear ratio can be easily changed between the input and output gears to achieve either torque or speed. The symmetric design would also allow for mounting a second set of legs on the other side, eliminating a complex crank shaft.

Why suitable: Our gear sketch verified gear alignment, rotation directions, and leg interference without having to make complex gears or build a difficult low-fidelity prototype.

2.3 PROTOTYPE SKETCHES

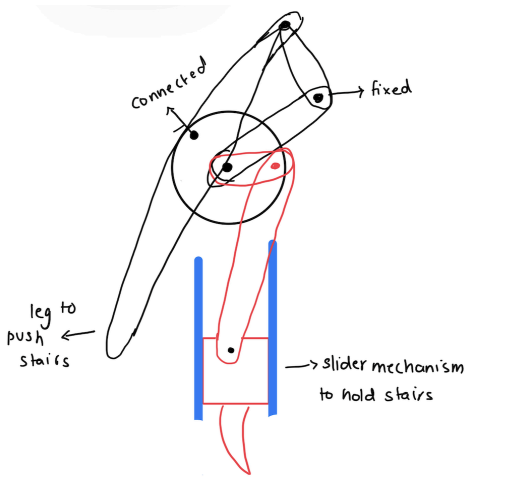


Fig 2. Sketch #1: answered Q2

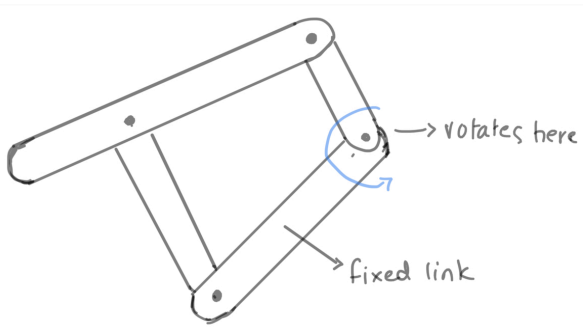


Fig 3. Sketch #2: answered Q1, Q2

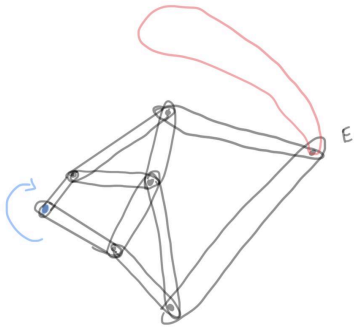


Fig 4. Sketch #3: answered Q2

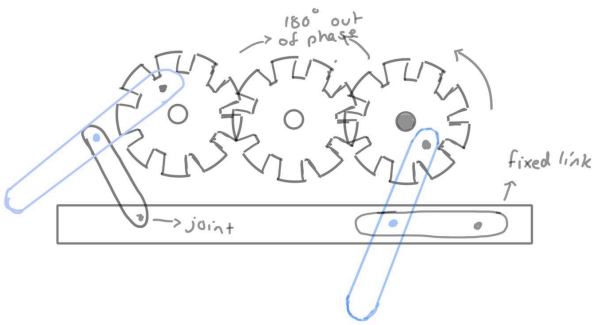


Fig 5. Sketch #4: answered Q3

2.4 PHYSICAL PROTOTYPE DOCUMENTATION

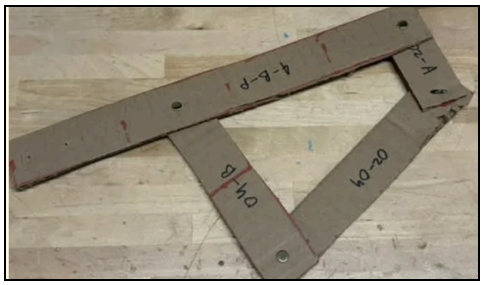


Fig 6. Low-fidelity prototype 1, discovered that small changes in link size creates large path changes

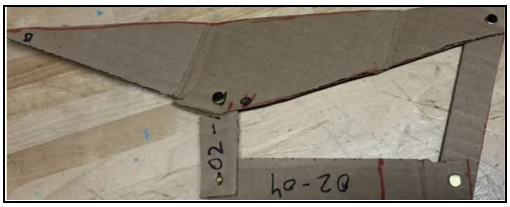


Fig 7. Low-fidelity prototype 2, discovered that misaligning links cause excess friction and link bending

2.5 REFLECTION FOR EACH PROTOTYPE

Since both our low-fidelity prototypes were four-bar linkages, testing different link lengths and link ratios whilst also noticing the wide variety of trajectories the mechanisms were achieving led us to focus more on trying to optimise a four-bar linkage for future design considerations. We did not completely eliminate six-bar linkages and crank slider mechanisms decided to prioritise four-bar linkages moving forward.

2.6 OUTCOMES

For future design and prototypes, we will ensure proper dimension transfer from our simulated models to our physical designs. We will also ensure linkage alignment: some of our fixed links and rotation links were not of appropriate sizes and interfered with the movement. We will also widen our joints to reduce the strain on smaller parts, and make our links chunkier for smoother rotation.

3. MODERATE-FIDELITY PROTOTYPING LAB

3.1 TOPOLOGY IDENTIFICATION

Sketch #1: 6 links | 7 joints | 6R + 1P | Ground: crank axle, rocker pivot and slider rails | Isomer: Watt (ternary crank disc and ground share the crank axle joint) | Crank-rocker leg plus crank-slider pawl on a common crank | Watt 2 with one prismatic joint	**Sketch #2:** 4 links | 4 joints | RRRR | Ground link connecting the two fixed pivots | Isomer: N/A (four-bar has only one connectivity) | Crank-Rocker | Four-Bar RRRR template
Sketch #3: 6 links (4 binary, 2 ternary: L3, L4) | 7 joints | 7R | Ground: L1 (binary, pivots at O2 and	**Sketch #4:** 8 links | 11 joints | 9 revolute + 2 gear meshes | Ground link carrying the three gear axles and both rocker pivots | Isomer: N/A (geared

O4) | Isomer: Watt (ternary L3 and L4 joined directly at J3) | Watt I six-bar | M = 1	linkage, outside the four-bar/six-bar templates) | Two crank-rocker four-bars driven 180° out of phase through a three-gear train | No builder template

3.2 MECHANISM SELECTION RATIONALE

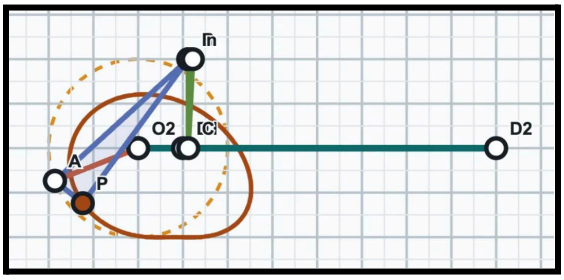


Fig 8. Kashvi

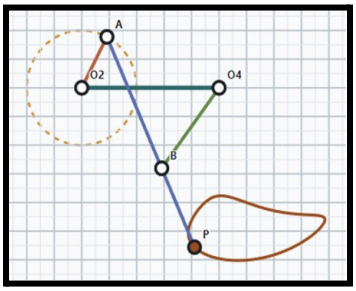


Fig 9. Will

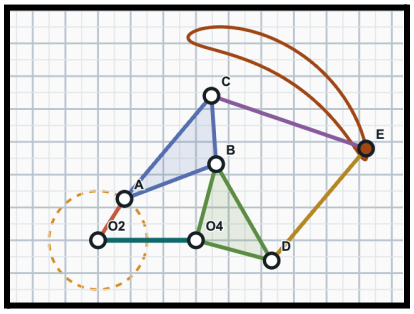


Fig 10. Kavish

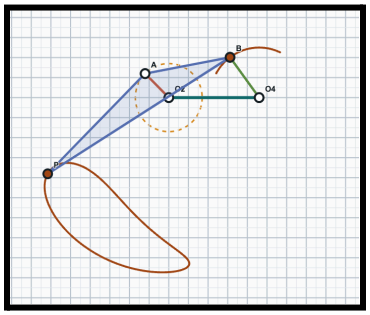
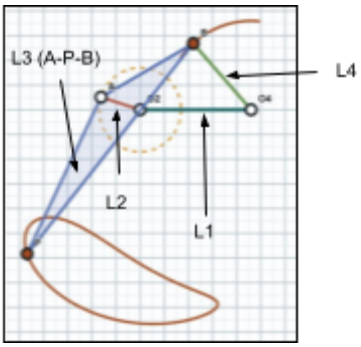


Fig 11. Daksh

The team considered four candidate mechanisms. Kashvi's crank-slider was simple but introduced friction and would require two separate slider assemblies for two legs, making it bulkier and harder to manufacture. Will's four-bar mechanism was also simple and reliable, but its slow return meant point P took too long to re-engage the stairs. Kavish's mechanism could generate the required motion, but it used too many links and followed an inefficient output path. Daksh's mechanism was selected because it provided a more efficient path for faster return, helping prevent unnecessary contact with the stairs. It was relatively simple to add a second leg operating out of phase for smoother, more stable motion.

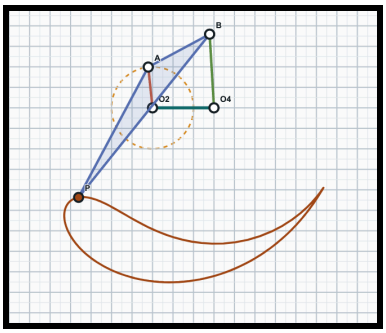
3.3 FINAL DIMENSIONS AND SIMULATION RESULTS

The final mechanism dimensions were L1 = 46 mm, L2 = 17 mm, L3 = 44 mm, and L4 = 36 mm, with point P at (-58 mm, -40 mm) relative to A. Simulation gave a 43 mm lift height and 80 mm stride length, satisfying the P1 requirements of at least 40 mm lift and 80 ± 5 mm stride. *Fig. 12* on the right.



3.4 ITERATION DOCUMENT

Before arriving at the design in Section 3.3, we used simulation to test different configurations and link lengths for Daksh's idea. In one failed configuration, the contact portion of the coupler curve was too curved, forcing the leg to change direction sharply, which could add joint load and cause jamming. We adjusted the link dimensions and pivot locations to flatten the bottom of the curve, so the foot pushed against the stair more consistently, and to incline the return path slightly, so the leg lifted clear of the stair without interference. This showed that both the contact and return portions of the path matter for smooth motion and reliable force transfer. *Fig. 13* above shows the failed trajectory.



3.5 FABRICATED MECHANISM AND MANUFACTURING DOCUMENTATION

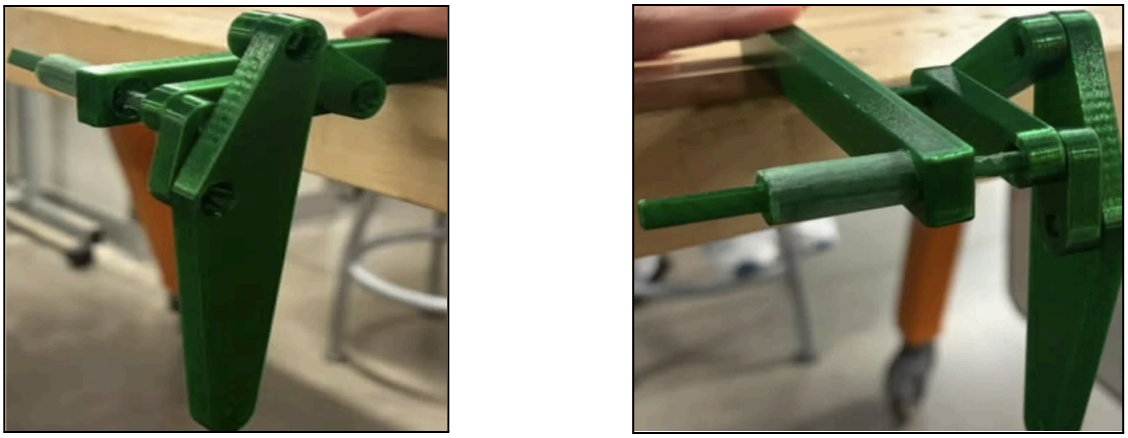


Fig 14. 3D Printed Four-bar linkage

For the moderate-fidelity prototype, we chose not to print the frame that would eventually hold the mechanism over the rack for the final design to avoid spending additional cost. We instead decided to come up with a compact bracket that can be held off the edge of a table or in our hands whilst still being able to help others understand the motion. The CAD model beside shows what the mechanism will potentially look like with a frame that is clamped to the ground and a handle to provide the input angular velocity.

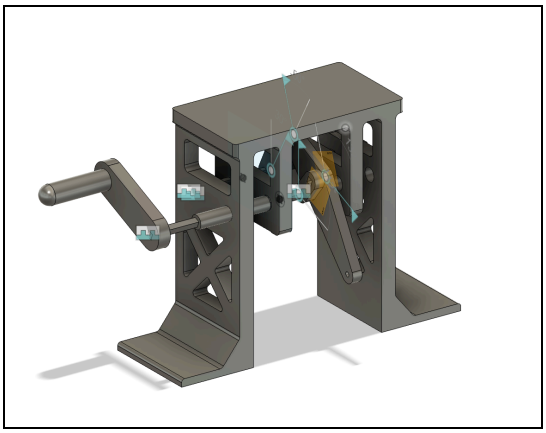


Fig. 15 Moderate-fidelity frame CAD model

3.6 REFLECTION

The moderate-fidelity prototype exposed several problems in our 3D printing process. First, the links and joints were too slender, which made the parts delicate, difficult to assemble, and hard to read as a mechanism. Second, some links slid out of plane as the crank turned, which showed that joint tolerances directly affect whether the linkage holds its motion, so we made joint fit a priority. Before the design review, we had assumed PLA joints were the main weakness because the material is brittle. Feedback from the review shifted our focus: increasing the joint diameters and making the links chunkier should come first, since sturdier geometry removes other possible failure causes. Only then could we judge whether the joint type or material was actually the problem. Because the link lengths stay the same, these changes strengthen the parts without altering the foot path.

4. APPENDIX - BUDGET UPDATE

Description	Qty/$	Unit Price	Qty Purchased	Source	Purpose	Tool Cost ($)
3D Printing (PLA filament)	1g/$0.03	0.03	39.48	Innovation Studio	Mid-Fidelity Protype	1.18
Manufacturing cost for printing	$0.025 per cm^3	0.025	71.4	Innovation Studio	Mid-Fidelity Protype	1.79
TOTAL						2.97